

Basic Properties of Continuous Functions

For any real number c , suppose the functions f and g both are continuous at $x=c$. Then, each of the following are continuous at $x=c$.

Constant Multiple

$$kf$$

Sum

$$f + g$$

Difference

$$f - g$$

Product

$$fg$$

Quotient

$$\frac{f}{g} \quad \underline{\text{provided}} \quad g(c) \neq 0$$

Algebraic Power

$$[f]^n$$

Composition

$$f \circ g$$